

Econometrics I

Lecture 0: Introduction

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Econometrics Overview

The aim of the course is to teach you to **use** popular applied econometric methods while developing your theoretical understanding of those methods. Topics include least squares, asymptotic theory, hypothesis testing, instrumental variables, difference-in-differences, regression discontinuity, treatment effects, panel data, maximum likelihood, and discrete choice models.

Prerequisites for Studying Econometrics

Ideally, you should have experience with

- ▶ Calculus
- ▶ Basic Probability and Statistics
- ▶ Linear Algebra
- ▶ Data analysis software such as R, Python/Numpy, Matlab, Julia, Stata, or similar (Excel doesn't count)

Rusty on linear algebra? Work through the self-study [Linear Algebra Review](#) (also linked on the course page) before Session 2. Khan Academy's *vectors and spaces* and *matrix transformations* units cover anything the handout skips.

Assignments and Grading

► 4 Problem sets: 10% each

- You may use any software you like. When giving examples, I will use R.
- To compose, consider using software like LaTeX, R Markdown, and Jupyter.

► Two in-class quizzes: 10% each

► Group project: 40%

- A research project on any topic (subject to my approval) using econometric analysis.
- I suggest choosing a published paper to replicate and finding a way to extend, test, or improve on it. Ideally, choose something on a topic that interests you. I will also share a list of suggestions.
- 1-3 students per group
- Proposal due in middle of semester.
- Presentations in final class session
- Paper due at end of semester

► Experiments and Research Design:

- In natural sciences, randomized controlled trials are considered the gold standard.
- In the social sciences, it's often hard to run experiments: macroeconomic policy, mergers and antitrust policy.
- This is perhaps the main reason we have econometrics: in the absence of controlled experiments, we need to figure out how to learn what we want to learn from naturally occurring data.

► Econometric Questions:

- Often about *causality*
 - About individuals: *What is the effect of education on wages?*
 - About markets (micro): *How does the price of the iPad affect the number of units that will be sold?*
 - About markets (macro): *How does raising the minimum wage affect employment?*
- Some studies are primarily descriptive
 - *How many home runs will Aaron Judge hit in 2025?*
 - *How has the distribution of real income in the US changed since 1990?*

What is Econometrics?

► Challenges:

- Endogeneity: general term for an observed variable's being correlated with things we can't observe.
Related: omitted variables.
- Selection: economic agents (people, firms, etc.) are purposeful and know more than we do about their personal situations!
- Simultaneity: is the relationship between price and quantity increasing (supply) or decreasing (demand)? What is driving the changes? The world is complicated, and it's often too simplistic to say *"this is the effect of X on Y"*.
- External validity:
 - What will happen if we raise the minimum wage to levels not before seen?
 - Is the price variation in the data short-run or long-run? Do consumers/firms respond differently to the two types of variation?
 - Related: structural econometrics, theory building.